



Indian Institute of Technology Hyderabad

Assignment 6 (Lab Report)

Basic Bioinformatics (BM2023)

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Question 1: Collect PDB files of proteins and visualize them in PyMOL.

- Worked with the GLUT-4 protein, I retrieved its PDB files using the 4-letter code from the RCS PDB website
- Visualized structures using PyMOL commands.
- Generated images showing overall tertiary structure.

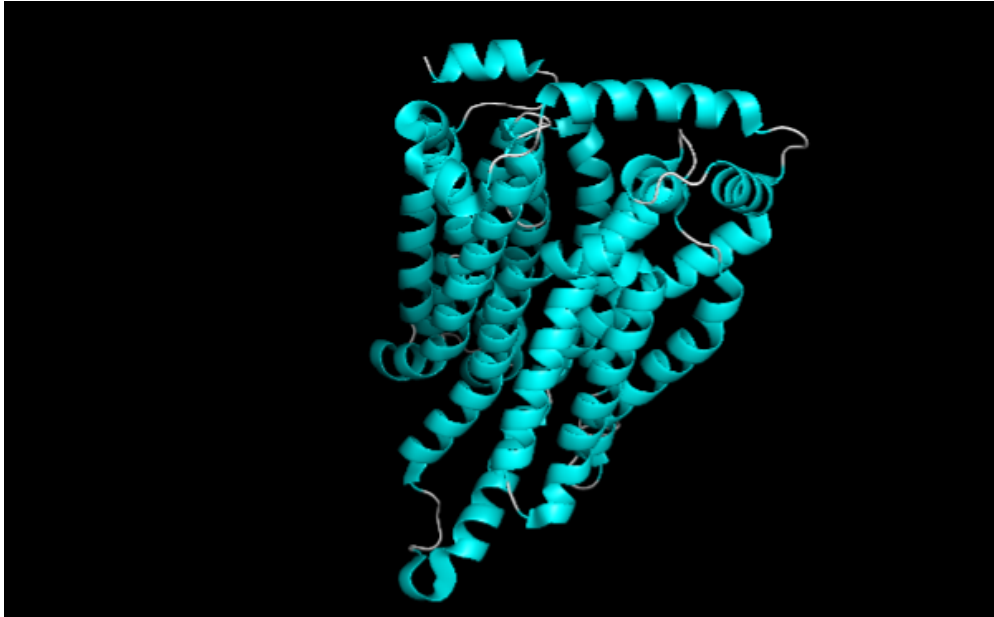


Figure 1: GLUT-4 protein visualized using Pymol

Question 2: Identify regions of interest using PyMOL.

Salt Bridges and Other Observations

During my analysis, one fascinating feature I came across was the presence of **salt bridges** in the protein structure.

These are **non-covalent interactions** that form between **side chains carrying opposite charges** in amino acid residues. They typically occur between:

- **Acidic residues:**
 - Aspartate (**Asp, D**) – Glutamate (**Glu, E**)
- **Basic residues:**
 - Arginine (**Arg, R**)
 - Lysine (**Lys, K**)
 - Histidine (**His, H**) (*depending on its protonation state*)

These salt bridges play a crucial role in maintaining the **three-dimensional structure** of proteins. They are especially important in densely packed regions, such as membrane proteins or active sites, where structural stability is vital. Using visualization tools, I was able to not only identify these bridges but also measure the distances between the interacting residues. Additionally, I explored another notable visualization method, the **dot representation** of the protein, which provided further insights into its spatial configuration.

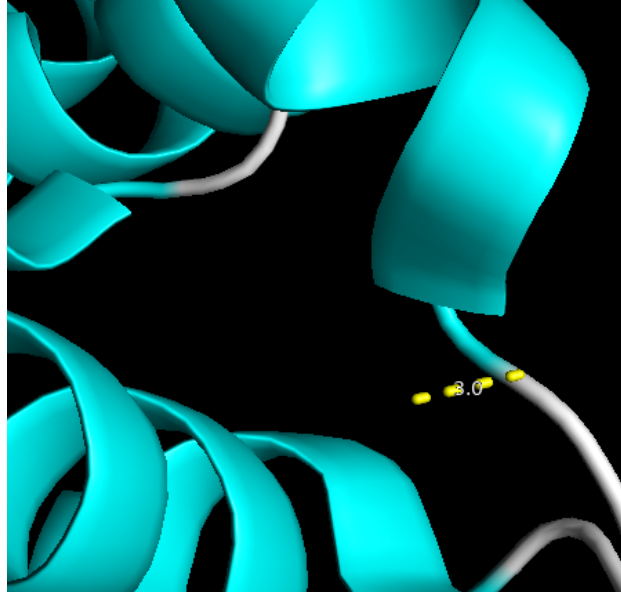


Figure 2: GLUT-4 protein visualized using Pymol

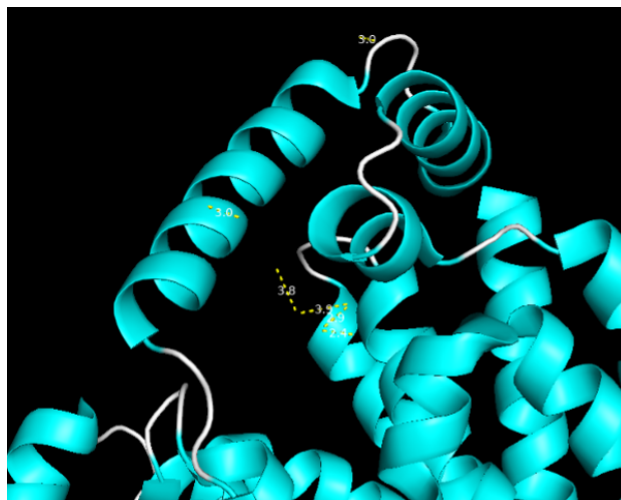


Figure 3: GLUT-4 protein visualized using Pymol

Another interesting thing I found is the ball and stick representation and we can also sort of dive into the protein for much detailed view.

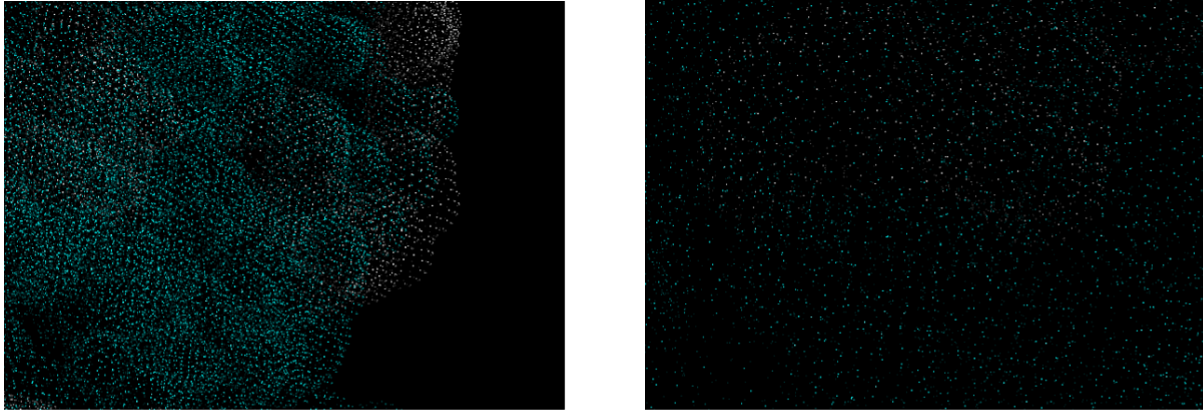


Figure 4: Each dot in the above image , represents each atom.

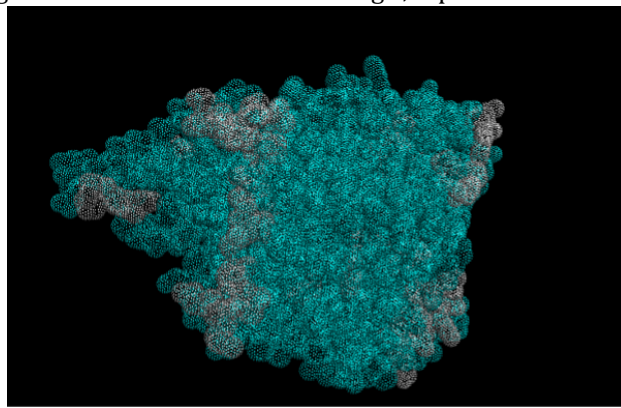


Figure 5: Each dot represents each atom , I found this very interesting

Question 3: Exploring Proteins from Various Classification Categories

- **Cellular Role:** GLUT4 plays a key part as an insulin-regulated glucose transporter, responsible for mediating glucose uptake in adipose tissues and muscles.
- **Family and Evolutionary Background:** This protein is a member of the Major Facilitator Superfamily (MFS), a large group of transporter proteins that are evolutionarily conserved across many organisms.
- **Functional Mechanism:** GLUT4 operates through facilitated diffusion, alternating between outward-facing and inward-facing conformations to enable glucose transfer across the plasma membrane.

Question 4: Protein-Receptor Structural Alignment using PyMOL

Although GLUT4 is often studied in relation to insulin signaling, it's important to note that **GLUT4 is not a hormone**. Unlike insulin, it **does not directly interact with a receptor** in the form of a ligand-receptor complex.

GLUT4 acts as a transporter that relocates to the **cell membrane** once the **insulin receptor** is triggered. While its action is **regulated by the receptor**, GLUT4 **does not bind** to it in the way a ligand typically would.

For comparison purposes, I aligned GLUT4 with insulin in PyMOL. However, this alignment is purely structural, as these two proteins do not form a complex in vivo.

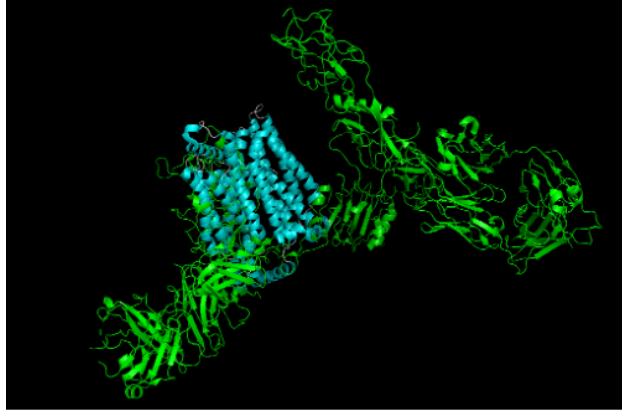


Figure 6: Protein-GLUT-4 , Receptor – Insulin (They don't form a complex)

Question 5: Investigating Structural Information through the CATH Database

What is CATH? The CATH database classifies protein domain structures based on their evolutionary relationships and structural organization.

While direct data for the GLUT4 protein is not available in the CATH database, I explored its structural family to derive relevant insights.

Considering GLUT4's structural properties and its association with the Major Facilitator Superfamily (MFS), the following structural features can be inferred:

- **Class:** Mostly Alpha – its structure is largely made up of alpha-helices.
- **Architecture:** Transmembrane Helical Bundle – helices span across the membrane in a bundled arrangement.
- **Topology:** MFS Fold – characterized by 12 transmembrane helices organized symmetrically into two domains.
- **Homologous Superfamily:** Major Facilitator Superfamily – a group of transporter proteins sharing evolutionary similarity.

These structural characteristics are commonly seen in MFS transporters, which are crucial for the movement of small solutes, such as glucose, across cellular membranes through facilitated transport mechanisms.